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Affective-CoT: Decomposing Multimodal Emotion Reasoning through a Hierarchical Cognitive Workflow

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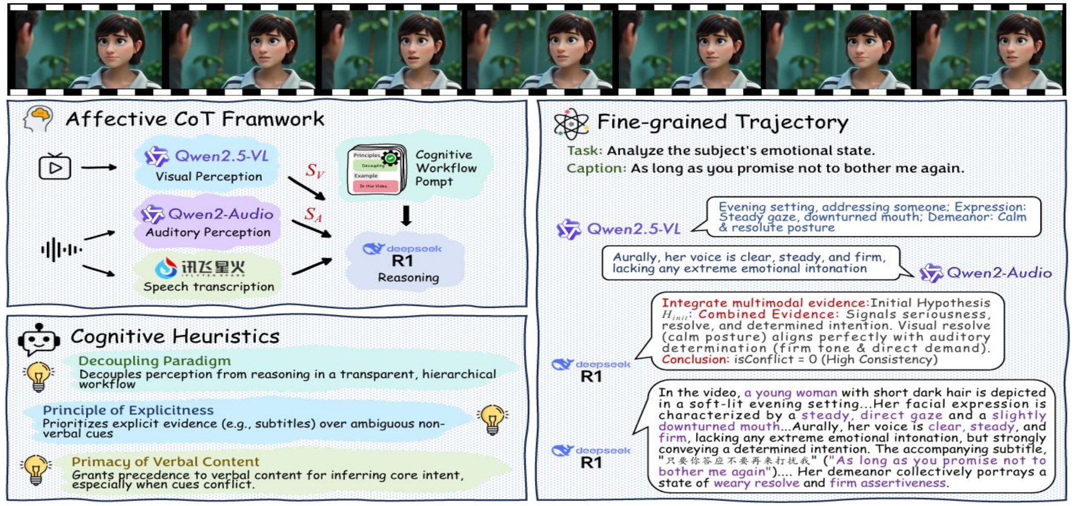
Framework

Affective-CoT introduces a hierarchical cognitive framework comprising three specialized stages—**Multimodal Evidence Perception**, **Cognitive Workflow Reasoning**, and **Descriptive Synthesis**——that decomposes emotion reasoning into interpretable steps to produce evidence-grounded, trustworthy emotional descriptions.

Method



Algorithm 1 Cognitive Workflow for Affective Reasoning

Require: Structured evidence S_V, S_A; Master prompt P_R.
Ensure: Core emotional hypothesis H_{emo}; Reasoning trace R_{trace}.

```
function CognitiveWorkflow(S_V, S_A, P_R)
  Step 1: Evidence Integration & Initial Assessment
    E_int ← Synthesize(S_V, S_A)
    H_init ← GenerateHypothesis(E_int, P_R)          ▷ LLM-based
    R_trace ← InitializeTrace(H_init)
  Step 2: Cross-Modal Consistency Analysis
    isConflict, C_points ← AnalyzeConsistency(E_int, P_R)
    if isConflict then
      R_trace ← AppendToTrace(R_trace, C_points)
   ↪ Step 3: Conflict Resolution
      H_res ← ResolveConflict(C_points, P_R)
      H_emo ← H_res
      R_trace ← AppendToTrace(R_trace, H_res)
    else
      H_emo ← H_init
    end if
    return H_emo, R_trace
end function
```

Experiments & Results

Type	Model	Method	FG-Score (↑)	EGS (↑)	CRR (↑)
Closed-Source	GPT-4o[19]	Few-Shot	0.5773	0.83	0.71
	Gemini 2.5 Flash-Lite[40]	Few-Shot	0.5823	0.89	0.82
Open-Source	AffectGPT[27]	Publication score[24]	0.4686*	-	-
	PandaGPT[38]	Few-Shot	0.5592	0.67	0.61
	Chat-UniVi [21]	Publication score[24]	0.3362*	0.57	0.37
	Video-LLaVA [31]	Publication score[24]	0.1979*	0.35	0.36
	MiniGPT4-Video [2]	Few-Shot	0.5119	0.67	0.63
	LLaVA-OneVision [22]	Few-Shot	0.5672	0.71	0.78
	Qwen2-VL [42]	Few-Shot	0.5684	0.78	0.65
	Qwen2.5-VL [3]	Few-Shot	0.5694	0.85	0.77
	InternVL2 [7]	Few-Shot	0.5719	0.82	0.78
	Video-LLaMA 2[9]	Few-Shot	0.5361	0.73	0.63
	LLaVA-NeXT[33]	Few-Shot	0.5702	0.84	0.72
Our Variants	Affective-CoT (InternVL-P[7])	Few-Shot CoT	0.5761	0.90	0.83
	Affective-CoT (SALMONN-P[39])	Few-Shot CoT	0.5717	0.88	0.81
	Affective-CoT (Llama3-R[14])	Few-Shot CoT	0.5786	0.91	0.85
	Affective-CoT (DeepSeek-V3[32])	Few-Shot	0.5793	0.91	0.85
	Affective-CoT (DeepSeek-R1[16])	Few-Shot CoT	**0.5805**	**0.92**	**0.86**

Model	Acc & W-F1 Avg.
Emotion-LLaMA[8]	0.7308
AffectGPT[27]	0.7497
Affective-CoT (w/o Workflow)	0.7595
Affective-CoT (Full)	**0.7600**

Model Type	FG-Score (↑)	EGS (↑)	CRR (↑)
Affective-CoT	**0.5805**	**0.92**	**0.86**
w/o Cognitive Workflow	0.5697	0.61	0.25
w/o Structured Evidence	0.5711	0.75	0.68

1st place in the MER-2025 challenge

$$(E_{desc}, Expl) = f_{R-LLM}(H_{emo}, R_{trace}, S_{VA}, P_R[\text{step4}])$$

$$(E_{desc}, Expl) = F_{CoT}(V, A, T)$$

Affective-CoT leverages specialized models to extract structured multi-modal evidence and employs a reasoning LLM to integrate information and generate interpretable descriptions. Our method achieves state-of-the-art performance in fine-grained emotion understanding (FG-Score: 0.5805) and significantly surpasses baselines in interpretability (EGS: 0.92, CRR: 0.86). Ablations confirm the necessity of both components.